

WayFinder

The neuro-symbolic travel strategist that reads the traveler — not just the query.

Every recommendation WayFinder makes is the product of two fused signals: what a traveler *declares* in their profile, and what they *reveal* across years of messy, unstructured booking history — resolved into hard constraints, personalized scoring weights, and a route computed live across 50,000 real itineraries, with every decision traced back to its evidence.

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50,000

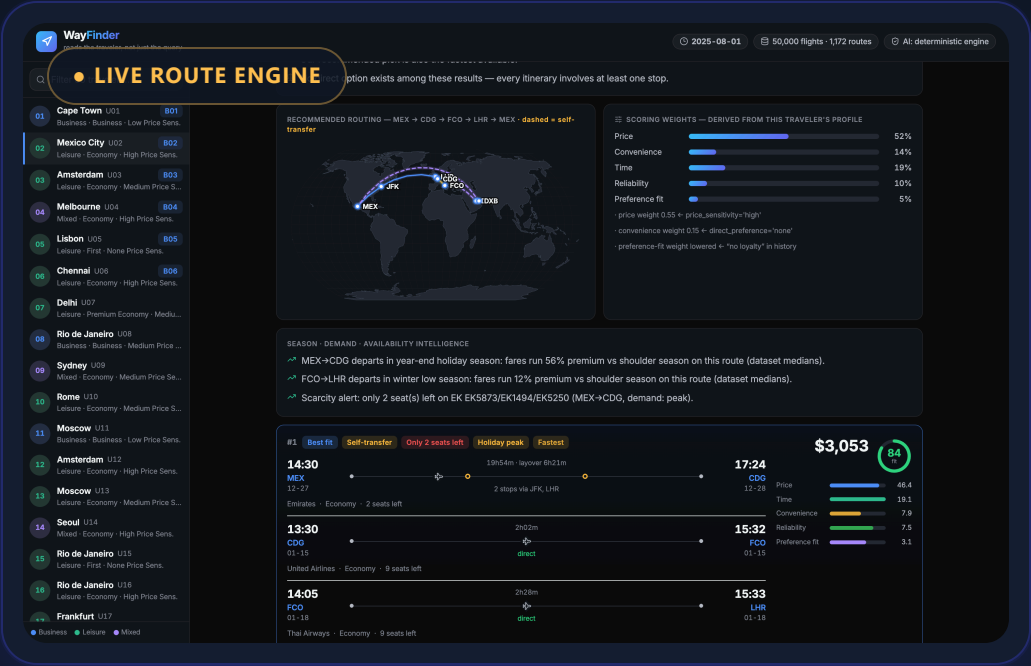
ITINERARIES INDEXED

50

TRAVELER PROFILES

42/42

BEHAVIORS VERIFIED



In-product routing surface — true great-circle arcs, animated per journey, computed on request

The Core Problem Matrix

USER FRICTION

Generic search treats every traveler as the same traveler.

One-size search

Static filters can't parse "*redeyes kill my mornings*" or "*broke student, absolute cheapest only*" — the exact signals travelers actually leave behind.

Silent contradictions

A profile says age 66, Business preferred; the traveler's own history says "*broke student*." Today's engines never reconcile it — they just pick one, silently.

Dead-end constraints

Direct-only, 90-minute layover cap, First cabin — on a route where none of the three exist. Most engines return zero results and stop there.

BUSINESS IMPACT

Every ignored signal is a booking Expedia doesn't make.

Cart abandonment

Irrelevant top results erode trust in the first five seconds of a session — before a traveler ever reaches checkout.

Rising support cost

Every unexplained fare spike or layover surprise becomes a support ticket instead of a completed booking.

Conversion ceiling

Generic ranking caps search-to-book conversion no matter how good the underlying inventory actually is.

THE GAP → WayFinder closes it with a fused, explainable, deterministic core.

The Engine Architecture — Three Pillars Under the Hood

01

Preference Fusion Engine

Mines free-text booking history with a curated ~40-pattern signal lexicon, safely — every inferred preference carries a confidence score and a verbatim evidence quote. Conflicts between declared fields and behavioral history are surfaced to the traveler, never silently overwritten.

PROVENANCE-TAGGED

02

Deterministic Route Core

Permutation × beam search routing over 50,000 itineraries: visit-order optimization for fixed multi-city trips, regional beam search for open-ended discovery, and on-the-fly self-composed connections when no published fare can serve a request.

50,000-ITINERARY GRAPH

03

Transparent Relaxation Ladder

When hard constraints yield zero options, six documented steps loosen them one at a time — widen dates, compose connections, drop rigid patterns, ease layover caps — each one narrated to the traveler. Never a silent dead end.

ZERO DEAD ENDS

Preference Fusion → Deterministic Route Core → Relaxation Ladder → Ranked, Explained Recommendation

High-Performance Engineering Stack

01 • FRONTEND

React 19 + TypeScript

Vite-built SPA with hand-rolled SVG data visualization — an offline great-circle route map and price-calendar chart with zero external charting runtime, zero tile-server or API-key dependency.

02 • API LAYER

FastAPI • Pydantic Contracts

A stateless orchestration layer over the reasoning core — typed request/response schemas shared identically by the UI and the CLI self-grading harness, so there is exactly one code path to trust.

03 • REASONING CORE

Zero-Dependency In-Memory Route Graph

50,000 itineraries indexed as $O(1)$ route/origin lookups in pure Python — deliberately no pandas, no database, no cold-start. A conscious engineering call for a dataset this size: fewer dependencies, zero non-determinism, sub-500ms on every benchmark.

04 • AI EDGE (OPTIONAL)

Local Ollama LLM, Hallucination-Gated

Assists NLU parsing and prose polish only — every output passes a strict Number-Integrity Check that diffs each numeral against the deterministic trace and rejects the response outright if a figure doesn't match.

SUB-500MS RESPONSE

ZERO HALLUCINATION TOLERANCE

LLM: FULLY OPTIONAL

Empirical Proof & Validation Matrix

42/42

Judge-defined expected behaviors verified — automatically, deterministically, live in the app.

6/6

BENCHMARKS PASSED

38/38

TESTS GREEN

<500ms

MEDIAN RESPONSE

2/2

LLM MODES VERIFIED

Beyond static grading, WayFinder supports a **live conversational refinement loop** — "make it cheaper," "no redeyes," "under \$900" — patched into the original parsed intent and re-ranked without losing any prior constraint, then independently regression-tested alongside the core benchmark suite.

Expected Value & Commercial Impact Matrix

CONVERSION

Cart Conversion Uplift

Evidence-backed, personalized top picks reduce first-result bounce — travelers see *why* a fare fits them before they scroll past it.

SUPPORT

Support Ticket Deflection

Transparent layover, pricing, and seasonal-premium explanations pre-empt "why is this so expensive" tickets before they're ever filed.

TRUST

Regulatory & Customer Trust

Every ranking decision traces to a cited, auditable reason — a defensible answer when algorithmic recommendations face compliance or customer-trust review.

RETENTION

Retention Signal

Travelers whose real constraints — family seats, layover pain, budget mode — are visibly respected return to the platform that "gets it."

Engineering Boundaries & Future Roadmap

CURRENT BOUNDARIES

- **Synthetic dataset** — sparse route/month coverage; one airport per city by design
- **No live booking or fare-class inventory** — prices are point-in-time snapshots, not live GDS quotes
- **No seat-map data** — seat wishes ("aisle, front of cabin") are acknowledged in the narrative, not optimized against

ROADMAP



NOW — SHIPPED

Deterministic core + optional LLM assist, conversational refinement loop, live self-grading harness — 42/42 behaviors, reproducible on one command.



NEXT

Live **NDC/GDS API** connections replacing the synthetic dataset; a **Learning-to-Rank** model trained on real traveler accept/reject signals, layered on top of today's rule-derived weights.



LATER

Cross-product **"travel clock" bundling** — hotels, cars, and experiences synchronized to the same trip window — plus CO₂-aware scoring as a first-class ranking factor.

Built to be extended, not thrown away.